5G Network Overview

Course objectives:

With the fifth generation, not surprisingly named "5G", 3GPP has once again shaken the technology landscape for mobile networks by introducing entirely new concepts to organize, develop and operate mobile networks. In addition, 3GPP has also provided new and exciting approaches to diversify services, enabling network operators to tap into business segments, far beyond the traditional roles of an MNO / MVNO or Service Provider.

Whether you're an aspiring Network Engineer, a Project- or Technical Manager, Marketing Professional or Executive or simply curious about the 5G Network in more detail, this course will equip you with the comprehensive foundational knowledge you need.

We'll dive into key concepts, starting with the 5G technology drivers and 5G standardization, differentiating a 5G network from its predecessors, followed by a closer look at new concepts such as Network Slicing and Multi-Access Edge Computing.

Participants will explore the functionality of Radio Access Network (RAN) and Core Network (CN), learn about network sharing and core security concepts in 5G.

Finally, we'll discuss various 5G operation scenarios and deployment options, including capacity and performance considerations, and provide an outlook into the future 6G network concepts.

Duration Instructional methods:

Option 1: On-Site Training: 4 days,
Classroom-style lecture using Power-point slides with interactive discussion,
Questions & Answers after each module.

Option 2: Virtual Classroom: 4 days total with 4 x 1.5 hours per day

Option 3: Virtual Classroom: 8 days total with 2 x 1.5 hours per day

Upon successful registration participants will receive dedicated LMS account credentials, allowing access to pre-training documents and eLearning material.
All participants will also receive comprehensive course materials.

Target Audience:

The course is addressed to technical and non-technical staff from the technology, marketing and management field, regulators and government officials, executives, requiring an overview about the 5G network concepts, architectures, standards, implementations and applications.

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Course Contents:

- Introduction to 5G
  - The ITU IMT-2020 framework for 5th Generation Networks
  - 5G technologies and objectives
  - Drivers for 5G Networks
  - 5G features and their benefits
  - Enhanced Mobile Broadband
  - Massive Machine-type Communication
  - Ultra-reliable Low-latency Communication
  - Fixed Mobile Broadband

- 5G network architecture, key components and their role
  - Introducing SDN and Virtualization
  - The 5G Radio Access Network
  - The 5G Core Network options: 5G EPC and 5GC
  - Network Slicing
  - Multi-Access Edge Computing (MEC)
  - SDN and Virtualization in 5G Networks
  - The 5G Transport Network
  - Integrated Access and Backhaul (IAB)
  - Examples and case studies

- 5G Radio Interface
  - 5G Radio Frequency aspects
  - New 5G Access Protocols
  - Radio Interface Channel Model
  - Physical Layer Procedures
  - 5G Radio Resource Control and its states (RRC)
  - advancedMIMO, Beamforming and mmWave operation
  - The Heterogenous Network
  - Introduction to 5G Network Planning and Optimization

- Security in 5G Networks
  - 5G Security Architecture
  - User Equipment (UE) security aspects
  - Access Network security aspects
  - Core Network security aspects
  - Transport Network and IP security aspects

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- Signaling Protocols in 5G and selected End-to-End Interworking procedures
  - The signaling protocols stack
  - The 5G network registration procedure
  - End-to-End connection establishment
  - Mobility procedures in 5G networks
  - Interworking between LTE (4G) and 5G networks
  - WiFi support in 5G

- 5G Network Operation
  - LTE and 5G: coexistence and transition
  - 5G Deployment and Rollout options
  - Capacity and Performance considerations
  - Examples of 5G Network deployments

- Outlook into future mobile networks
  - Open RAN and Core Network Architectures
  - Vision towards 6G and 7G
  - Impact of Cloud, Quantum Computing and Quantum Communication to Mobile Networks
  - Introduction to Autonomous Networks (Self-X)

Pre-requisites:

Basic knowledge about mobile networks, IP networks and experience with IT infrastructure are beneficial.

Course Code: 5GSysO